

A Major Project Presentation on Space Weather Prediction Platform (CosmoPredict)



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Introduction

CosmoPredict: 3D Space Weather Forecast & Rocket Simulation



- Predicts solar flares and geomagnetic storms using Machine Learning.
- Combines historical and real-time space data from NASA, NOAA, and ACE satellites.
- Provides a futuristic **interactive dashboard** with 3D rocket trajectory simulation.
- Enhances safety for satellites, astronauts, and space missions.

Problem Statement

Space weather events like solar flares and geomagnetic storms pose serious risks to satellites, astronauts, and communication systems. Current monitoring systems collect vast data but lack predictive accuracy and real-time alerts. **CosmoPredict** addresses this gap by integrating AI and machine learning to forecast space weather events and provide actionable insights for timely preventive measures.

Unpredictable Events

Space weather events are **unpredictable**, affecting satellites, GPS, aviation, and power grids.

No Centralized Platform

Lack of **centralized, real-time monitoring platforms**.

Missing Alerts

Absence of **early warning alerts** for astronauts and space agencies.

Limited Guidance

Limited **operational guidance** for satellite trajectory adjustments during severe events.

Objectives

ML-Powered Predictions

Predict future solar flares and geomagnetic storms using **LSTM/GRU ML models**.

Real-Time Monitoring

Provide **real-time monitoring and interactive visualizations** on a web dashboard.

Automated Alerts

Generate **automated alerts** for high-risk events (dashboard, email, SMS).

Route Diversion

Introduce a **Route Diversion Module** to suggest safe orbital paths.

Educational Platform

Deliver an **educational and research-friendly platform** for space weather study.

Our AI-Driven Solution

Predictive AI

LSTM/GRU models
forecast solar activity



Automated Alerts

Email, SMS, and
dashboard warnings



Real-Time Monitor

Interactive dashboard
with live feeds



Route Diversion

Suggest safe orbital
adjustments

System Features



Real-Time Monitoring

Real-time space weather monitoring.



Predictive Analytics

Predictive analytics with historical data comparison.



3D Simulation

3D interactive rocket simulation for multiple deviation paths.



Alert System

Alert notifications via dashboard, email, and SMS.



Data Access

Historical data query and CSV/JSON download.



Responsive UI

Responsive UI with **dark/light mode toggle**.

Tools & Technologies



Frontend

React JS, Three.js, Tailwind CSS, Plotly/Chart.js.



Backend

Django REST Framework, PostgreSQL database.



ML Frameworks

TensorFlow, PyTorch.



Data Sources

NASA OMNIWeb, NOAA SWPC, ACE Satellite Data.



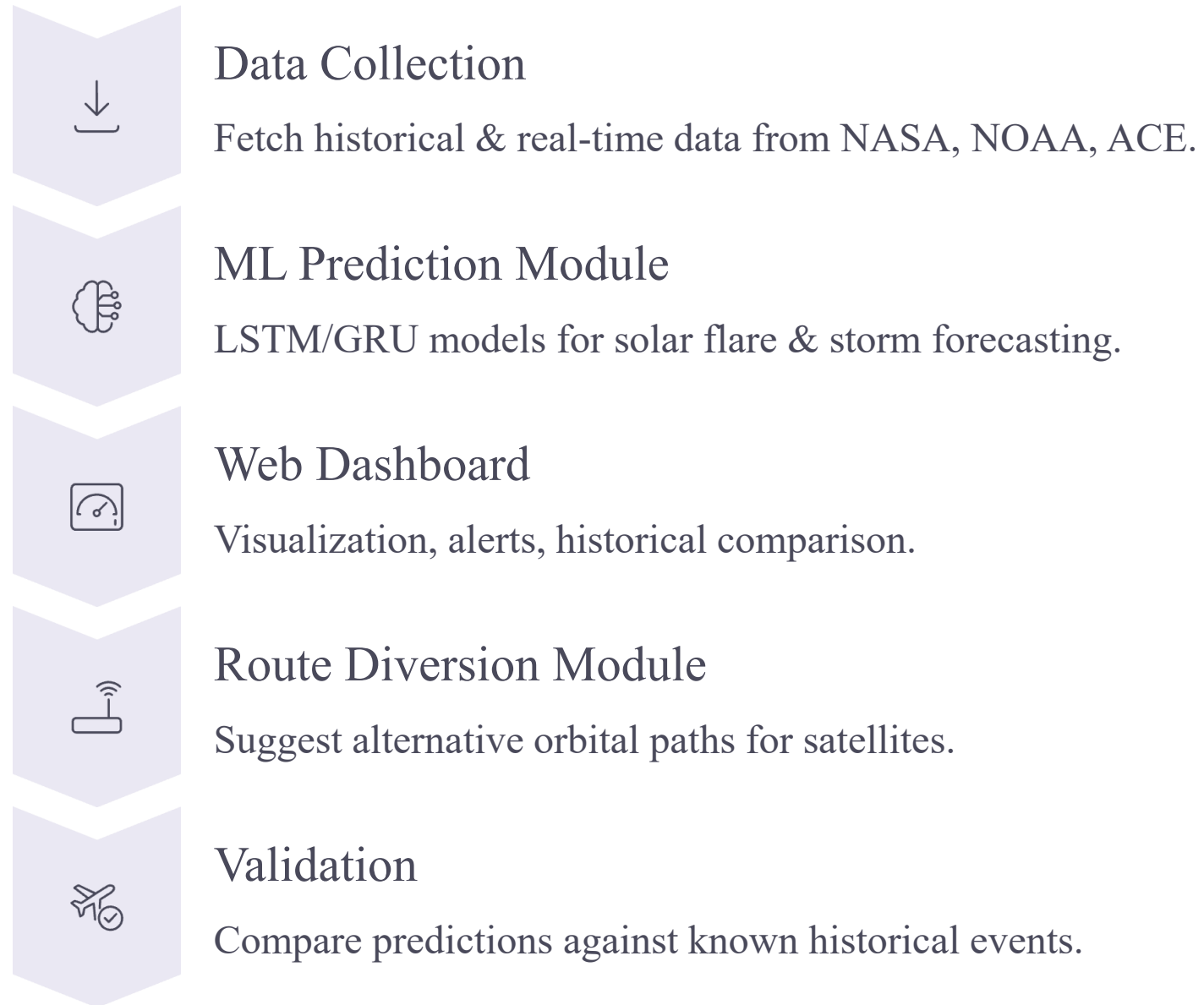
Others

GitHub for version control, cloud servers for hosting (AWS/GCP/Render).

System Architecture



A comprehensive workflow from data collection to actionable insights.



ER Diagram

Entities

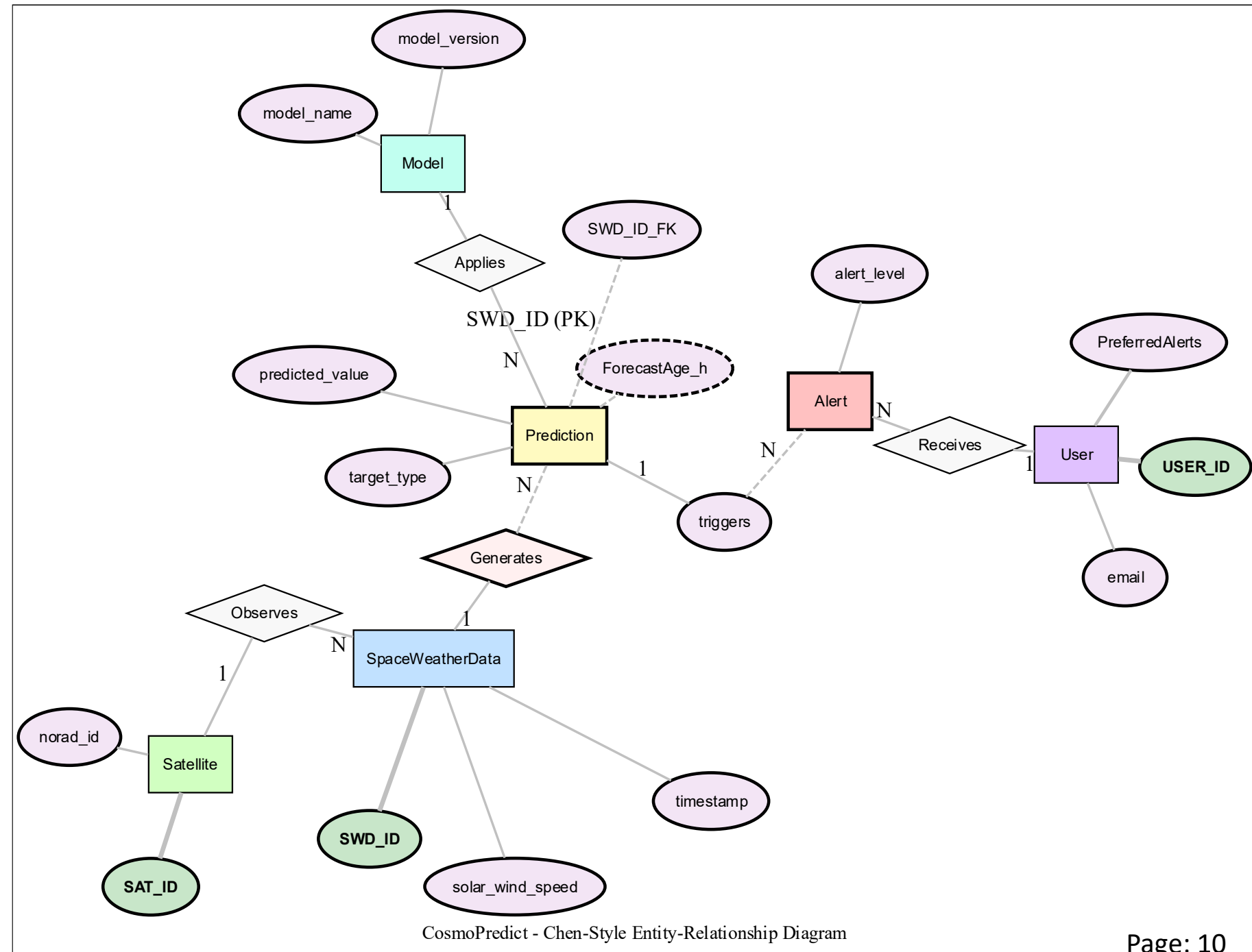
Space Weather Data, Rocket Trajectory, Users, Alerts.

Relationships

- Space Weather Data → Prediction → Alerts
- Rocket Trajectory → User → Alerts

Database

PostgreSQL storing processed and historical data.



UI Mockup: Dashboard



A comprehensive interface for monitoring space weather in real-time.



Real-Time Metrics

Real-time **space weather metrics**: solar wind, proton density, Kp index.



Interactive Charts

Interactive charts for **historical vs predicted events**.



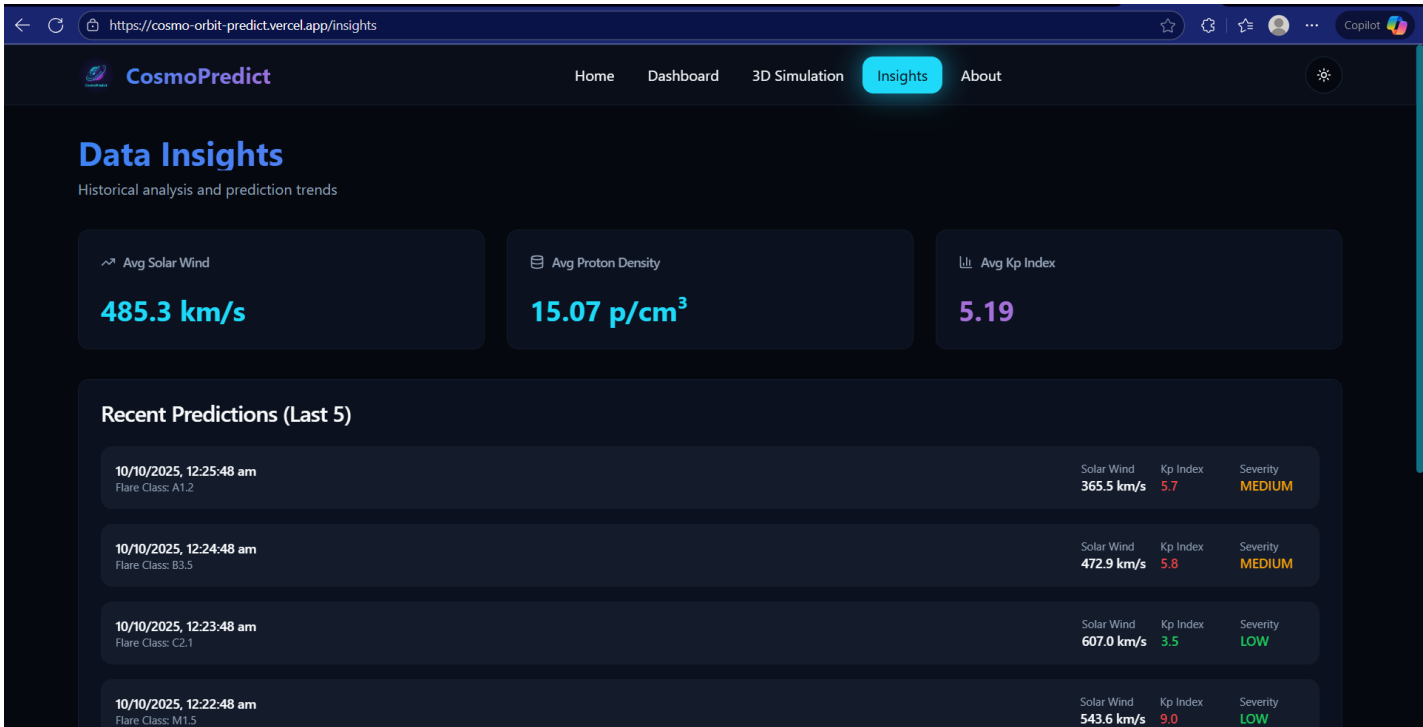
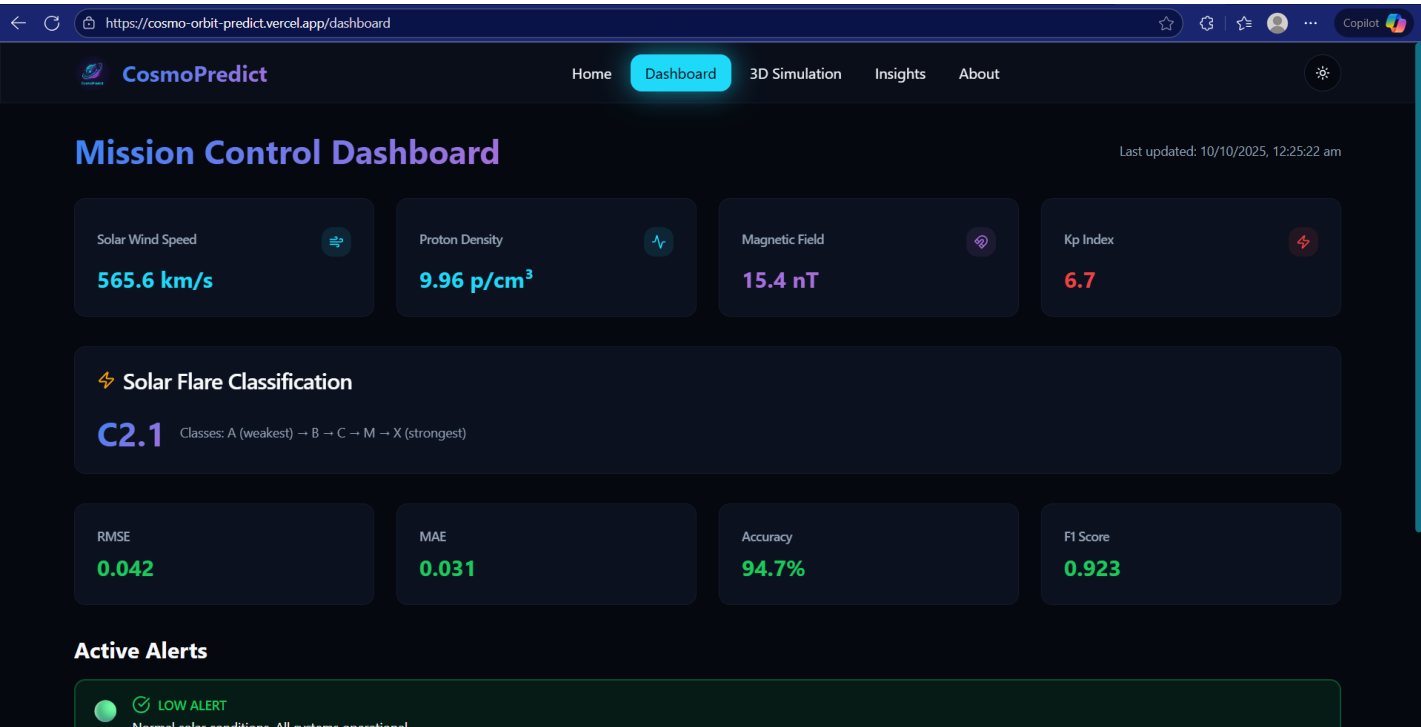
Color-Coded Alerts

Color-coded **alerts** for quick risk assessment.



Theme Toggle

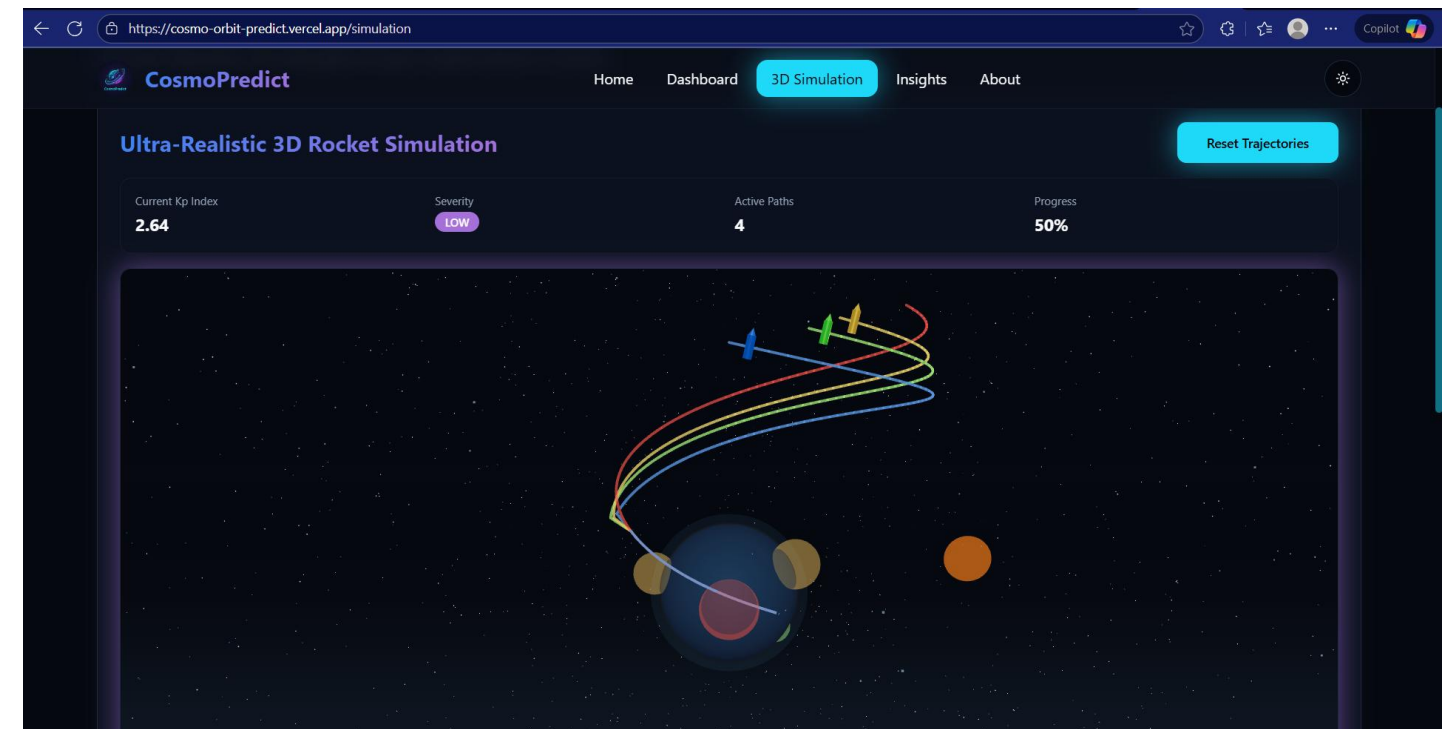
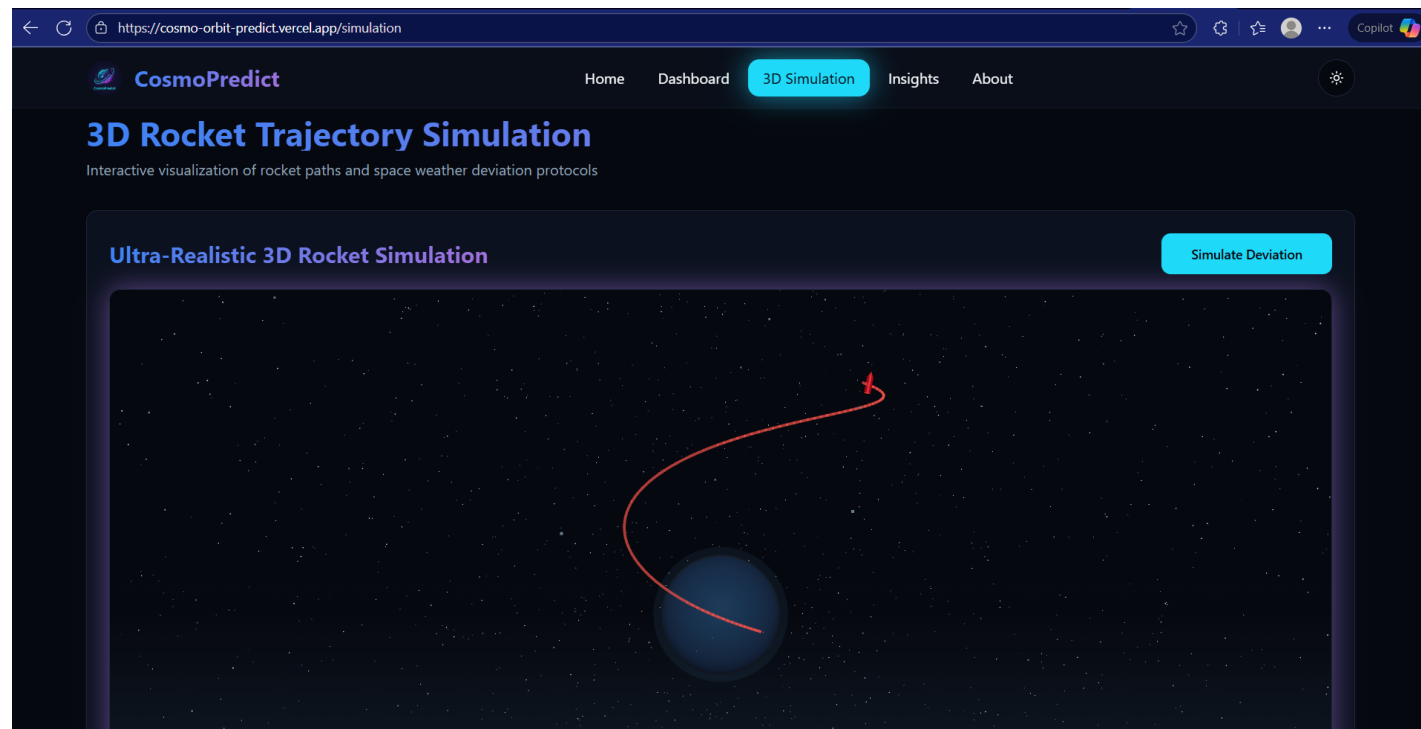
Light/Dark mode toggle for better usability.



UI Mockup: 3D Rocket Simulation

Interactive Trajectory Visualization

- 3D Earth globe with satellite and rocket trajectories.
- Multiple path simulations: **original trajectory (red)**, deviation paths (green, yellow, blue).
- Highlighted **affected zones** based on predicted space weather.
- User interaction: rotate, zoom, pan, and simulate deviation.



Key Advantages



Proactive Protection

Proactive protection for satellites and astronauts.



Real-Time Monitoring

Centralized real-time monitoring platform.



Predictive ML Models

Predictive ML models for accurate forecasts.



Operational Support

Operational support with orbital deviation suggestions.



Research & Education

Research & education tool for space science enthusiasts.

Future Scope



Multi-Satellite Missions

Extend predictions for multi-satellite missions.



AI Anomaly Detection

Integrate AI anomaly detection for solar wind and geomagnetic data.



Mobile Alerts

Mobile app for real-time alerts on-the-go.



Enhanced 3D Simulation

Enhanced 3D simulation realism with particle effects and trajectory analytics.



Solar Flare Classification

Incorporate solar flare severity classification and risk analytics.

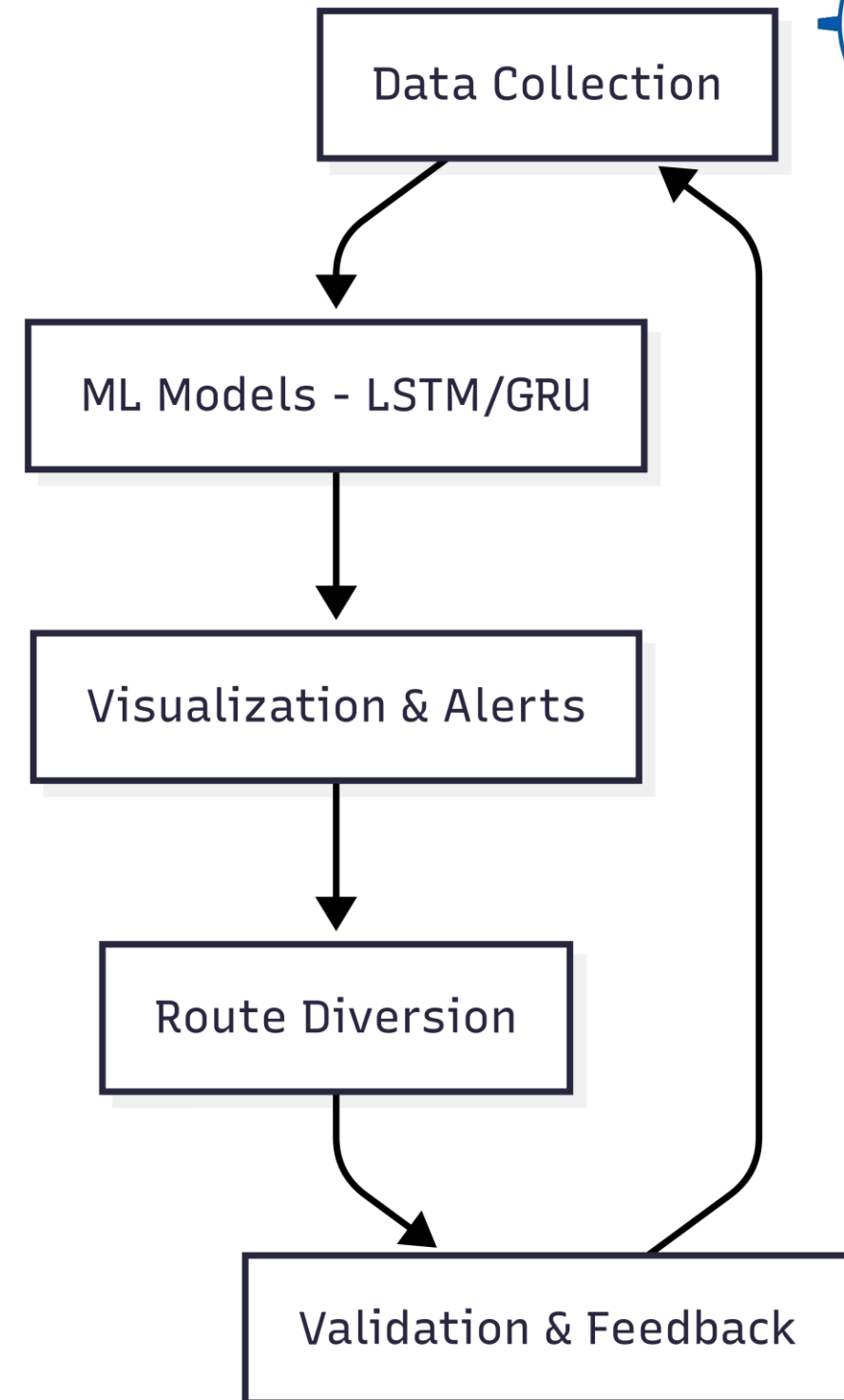
Feasibility & Scalability

Technical Foundation

- Cloud hosting ensures **scalable performance** for multiple users.
- Open-source ML frameworks and APIs reduce **development cost**.

Architecture & Performance

- Modular architecture allows **easy integration** of new data sources.
- Performance metrics ensure **real-time predictions under 5 seconds**.



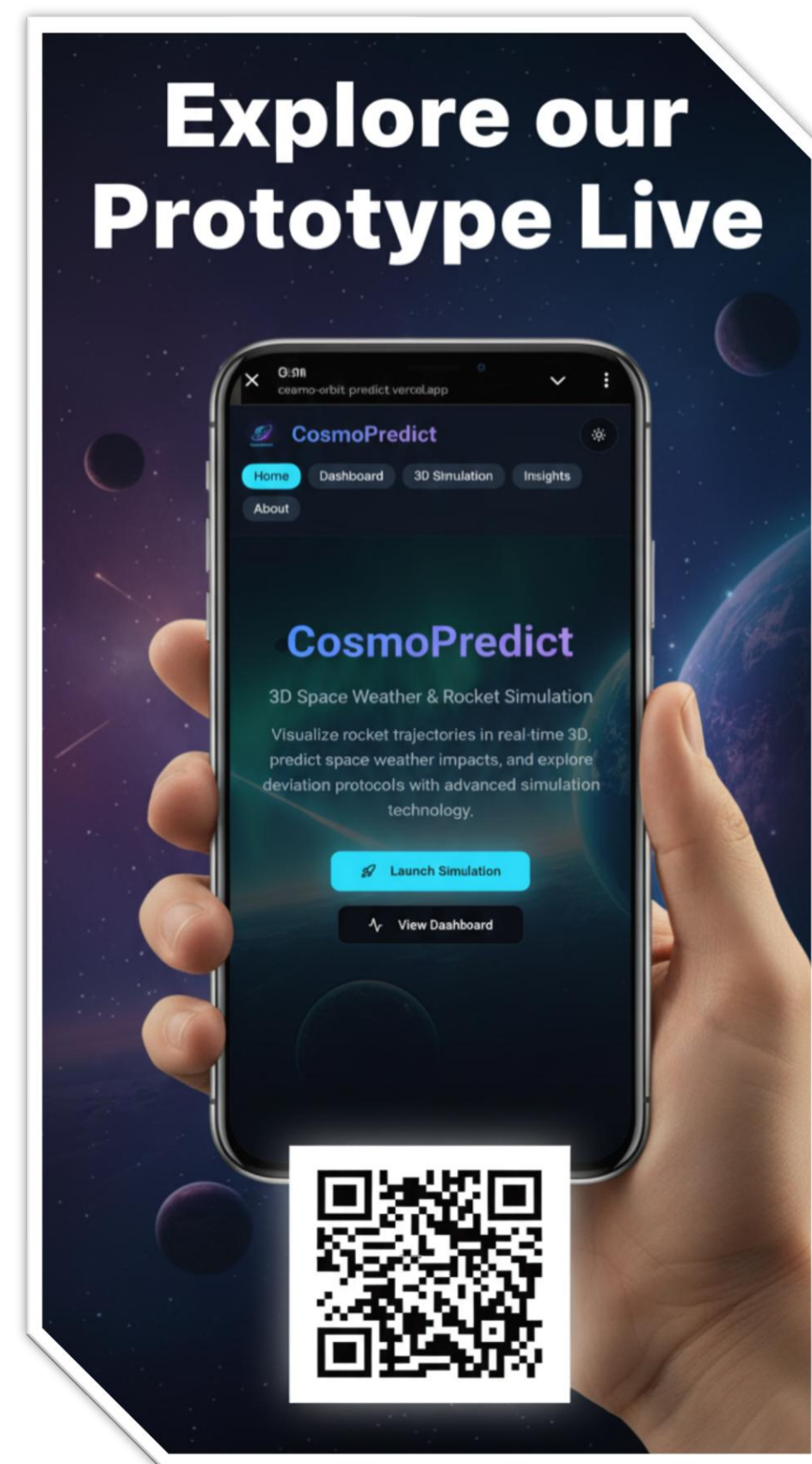
Conclusion

CosmoPredict combines **ML, real-time data, and 3D visualization** for space weather prediction.

Enhances **satellite safety, astronaut protection, and mission planning.**

Acts as a **research and educational tool** for space weather studies.

"We are confident in completing all tasks successfully and seek approval for project execution."





NASA OMNIWeb

<https://omniweb.gsfc.nasa.gov>

NOAA Space Weather

<https://www.swpc.noaa.gov>

ACE Satellite Data

<https://www.srl.caltech.edu/ACE/ASC/level2/>

Standards

IEEE Std 830-1998 (Software Requirements Specification Standard)

Documentation

Relevant ML & Python documentation




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Thursday, October 09, 2025 19:22:39 UTC

SPACE WEATHER CONDITIONS on NOAA Scales

24-Hour Observed Maximums

R1	S	G
minor	none	none

Latest Observed

R	S	G
none	none	none

Predicted 2025-10-09 UTC

R1-R2	25%	S1 or greater	5%	G	none
R3-R5	5%				

Solar Wind Speed: **330** km/sec

Solar Wind Magnetic Fields: Bt **7** nT, Bz **-6** nT

Noon 10.7cm Radio Flux: **120** sf

THIS SITE WILL REMAIN UPDATED DURING SHUTDOWN
published: Thursday, October 02, 2025 15:56 UTC
 Because the information this website provides is necessary to protect life and property, this site will be updated and maintained during the federal government shutdown.

Space Weather Update

SWFO-L1 Successfully Launched!
published: Wednesday, September 24, 2025 14:08 UTC
 SWFO-L1 successfully launched at 7:32am EDT on Wednesday, 24 September, 2025!

Artemis II Testbed Exercise Showcases NOAA's Readiness for Human Spaceflight
published: Monday, July 07, 2025 18:46 UTC
 The inaugural use of SWPC's Space Weather Prediction Testbed took place in spring 2025 during a multi-agency exercise preparing for NASA's Artemis II mission.

After Action Report of the 2024 Space Weather Simulation Exercise
published: Thursday, April 17, 2025 14:06 UTC
 The Johns Hopkins Applied Physics Laboratory released the results of the nation's first end-to-end Space Weather Tabletop Exercise (TTX).

ACE Level 2 (Verified) Data

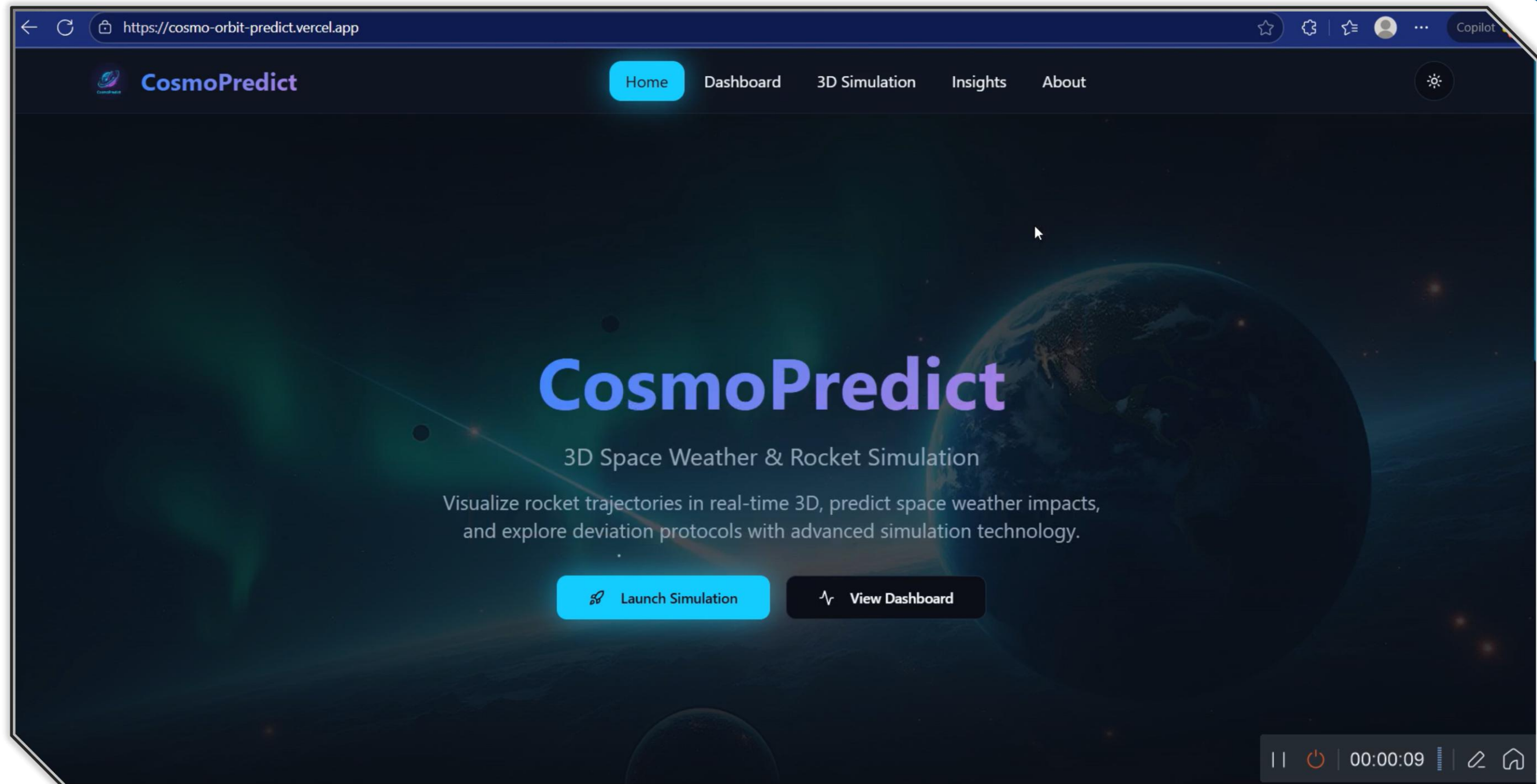
You might like to try our [alternate ACE Level 2 data server](#)

[ACE Level 2 Data Policy](#)

Interplanetary Magnetic Field Parameters: MAG.	MAG Data	Documentation
Solar Wind Parameters: SWEPAM	SWEPAM Data	Documentation
Solar Wind Temperatures, Speeds, Composition, Charge States: SWICS prior to August 23 2011	SWICS 1.1 Data	Documentation
Solar Wind Temperatures, Speeds, Composition, Charge States: SWICS after August 23 2011	SWICS 2.0 Data	Documentation
Solar Wind Proton density and speed: SWICS	SWICS Proton Data	Documentation
Solar Suprathermal and Energetic Particle Intensities: ULEIS	ULEIS Data	Documentation
Solar Energetic Particle Intensities: EPAM	EPAM Data	Documentation
Solar Energetic Particle Intensities: SEPICA	SEPICA Data	Documentation
SEP, GCR, and ACR Intensities: SIS	SIS Data	Documentation
Galactic Cosmic Ray Intensities: CRIS	CRIS Data	Documentation
Merged IMF and Solar Wind 64-second and Hourly Averages	MAG/SWEPAM Data	Documentation
Merged IMF, Solar Wind, and Energetic Particle Hourly Averages	Multi-instrument Data	Documentation

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Demo / Simulation Flow



Impact & Applications

Highlight potential use cases: satellite operators, space agencies, astronaut missions, research labs.



Satellite Operators

Real-time protection and operational guidance for commercial and government satellite fleets.



Space Agencies

Mission planning support and space weather forecasting for exploration programs.



Astronaut Missions

Safety monitoring and radiation exposure predictions for crewed space missions.



Research Labs

Educational platform and research tool for space weather studies and analysis.

